Problem Set M

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Learning the coefficients (β=1)

A monopolist have two objectives;

- What is the true profit maximizing price?
- Find the optimal price in a shortest possible time.

When there is no discount, a monopolist's priority would be the first objective. He would want to spend long enough time to estimate the demand curve carefully before he can settle in at a optimal price. But at what time should he stop the initial learning stage? How to design price points for efficient estimation of coefficients in demand curve?

$\Rightarrow$ Equally spaced price points should help efficient learning.

Design of initial learning stage prices

				12			
			6		18		
STEP1:		(6,12)			(12,18)		
STEP2:	(6,9)	(9,12)		(12,15)		(15,18)	
STEP3:	(6,7.5)	(7.5,9)	(9,10.5)	(10.5,12)	(12,13.5)	(13.5,15)	(15,16.5)	(16.5,18)

$$\vdots$$

After N steps, we have

$$3 + 2 + 4 + 8 + ... + 2^N = 3 + (2^{N+1} - 2) = 2^{N+1} + 1$$

price points.

STEP0 Let $p_1 = 12$, $p_2 = 6$, and $p_3 = 18$. Calculate coefficients in the demand equation b_{00}, b_{10}, b_{20}.

STEP1 Let p_4 be the middle point of (p_2, p_1) and p_5 be the middle point of (p_1, p_3). Calculate b_{01}, b_{11}, b_{21}. If

$$|(b_{00}, b_{10}, b_{20}) - (b_{00}, b_{10}, b_{20})| < \lambda,$$

stop the procedure, take these b_{i1}'s as the true coefficients. Otherwise, keep on the next step.

STEPN Let the next 2^N prices to be located at the middle of these intervals. Calculate coefficients in the demand equation b_{0N}, b_{1N}, and b_{2N}. If

$$|(b_{0(N-1)}, b_{1(N-1)}, b_{2(N-1)}) - (b_{0N}, b_{1N}, b_{2N})| < \lambda,$$

stop the procedure, take these b_{iN}'s as the true coefficients. Otherwise, keep on the next step until we reach the convergence.

STEP.. Once convergence of coefficient is achieved, calculate the expected profit maximizing price p_e using these coefficients. Let p_e be the price for all periods afterwards.

Trade off between maximizing current profit and learning

If $\beta < 1$, then a monopolist also need to focus on maximizing current profit, because not current profit means more than future profits. Being simple minded we may try to maximize current profit each time using available information only at that time.

Say we are at period $t \geq k+1$ for some $k \geq 3$. We know $p_1, ..., p_{t-1}$ and $x_1, ..., x_{t-1}$. We would find the optimal p utilizing all information we have, and choose this optimal p as p_t. But how to choose k and p_t's for $t < k$?

- The smaller the better k is.
- k should be at least 3.
- It is better if initial prices are equally spaced over possible price range for efficient coefficient estimation.

$\Rightarrow$ Choose k=3 and $(p_1, ..., p_k) = (12, 6, 18)$

Pricing A: optimize current profit at each period

A1. Let $(p_1, p_2, p_3) = (12, 6, 18)$.

A2. At each t for $t = 4, 5, 6, ...$, choose the p_t as following;

(i) Let $p = (p_1, ..., p_{t-1})'$ and $x = (x_1, ..., x_{t-1})'$. Run regression

$$\log(x) = b_0 + b_1 \log(p) + b_2(\log(p))^2$$

to get the coefficients b_0, b_1, and b_2.

(ii) Let $u = 50 : 180/10 = (5, 5.1, 5.2, ..., 18)$ and get the predicted values g of corresponding demand from the regression in (i).

$$g = \exp(b_0 + b_1 \log(u) + b_2(\log(u))^2$$

(iii) Calculate the profit for each u

$$profit = g'u - C(u)$$

(iv) Find the maximum profit and the corresponding u. Set this profit maximizing u as current period's price p_t. After some reasonable number of periods, this p_t will converge.

100 replications

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
6.6	8.6	9.4	9.18	9.83	11.4

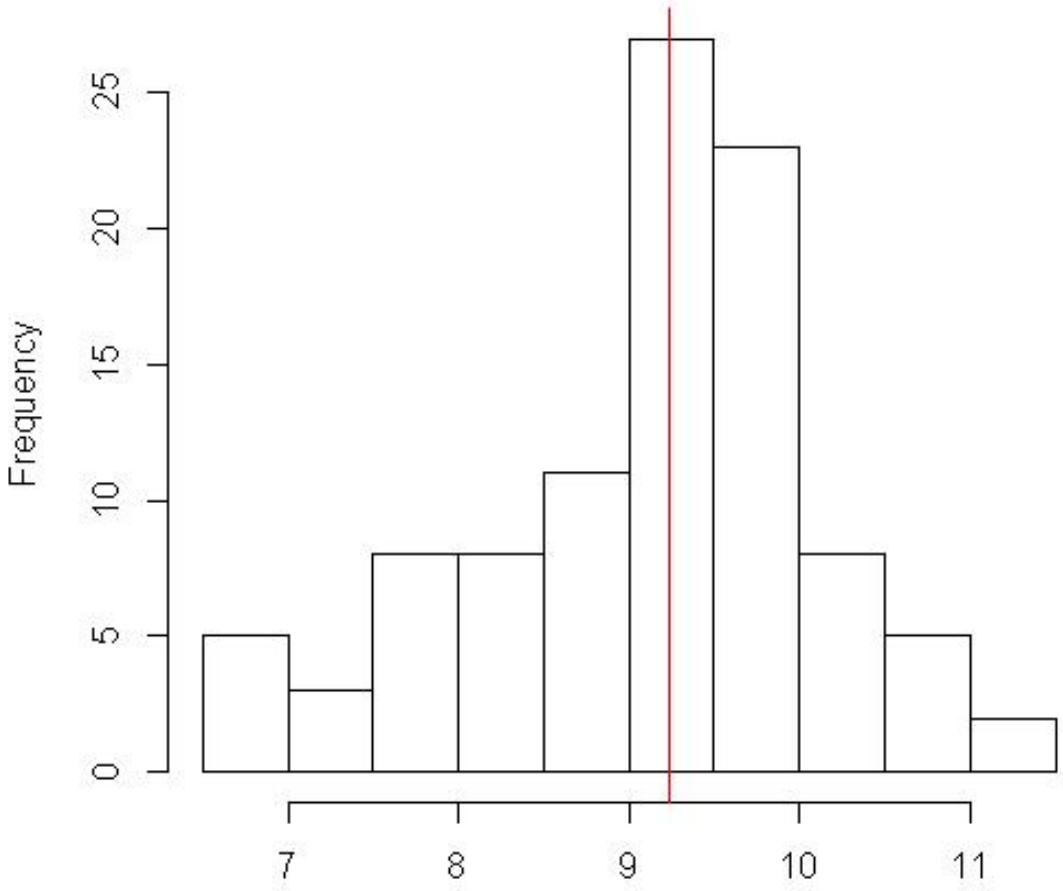

Figure: Distribution of optimal p's chosen by Pricing A: Red vertical line indicates the true optimal price 9.24

After a few periods (usually less than 10 periods) p_t's stay at the same points. Pricing Strategy A is risky in the sense that we may stuck at a price which is 'quite' far away from the true optimal price if we get unlucky at the first few periods.

What if we also care about coefficient learning? We would want to investigate at more wider range of prices.

$\Rightarrow$ Borrowing an idea from KW-procedure, give perturbation with $h_t = 3\beta t^{-1/6}$.

Pricing B: perturbation

B1. Choose some initial k prices $p_1,...,p_k$.

B2. At each t for $t = k+1, k+4, k+7, ...$, choose the p_t, p_{t+1}, and p_{t+2} as following;

(i) Let $p = (p_1, ..., p_{t-1})'$ and $x = (x_1, ..., x_{t-1})'$. Run regression

$$\log(x) = b_0 + b_1 \log(p) + b_2(\log(p))^2$$

to get the coefficients b_0, b_1, and b_2.

(ii) Let $u = 50 : 180/10 = (5, 5.1, 5.2, ..., 18)$ and get the predicted values g of corresponding demand from the regression in (i).

$$g = \exp(b_0 + b_1 \log(u) + b_2(\log(u)))^2$$

(iii) Calculate the profit for each u

$$profit = g'u - C(u)$$

(iv) Find the maximum profit and the corresponding $\hat{u}_{maxprofit}$ and let $p_t = \hat{u}_{maxprofit}$.

(v) Let $p_{t+1} = p_t - h_t$ and $p_{t+2} = p_t + h_t$.

B3. After some reasonably small number N of periods, the profit maximizing prices will be stabilized. Use the converged value p_e as the price for the later periods; $p_N = p_{N+1} = ... = p_e$.

100 replications

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
8.7	9.0	9.3	9.26	9.5	10.0

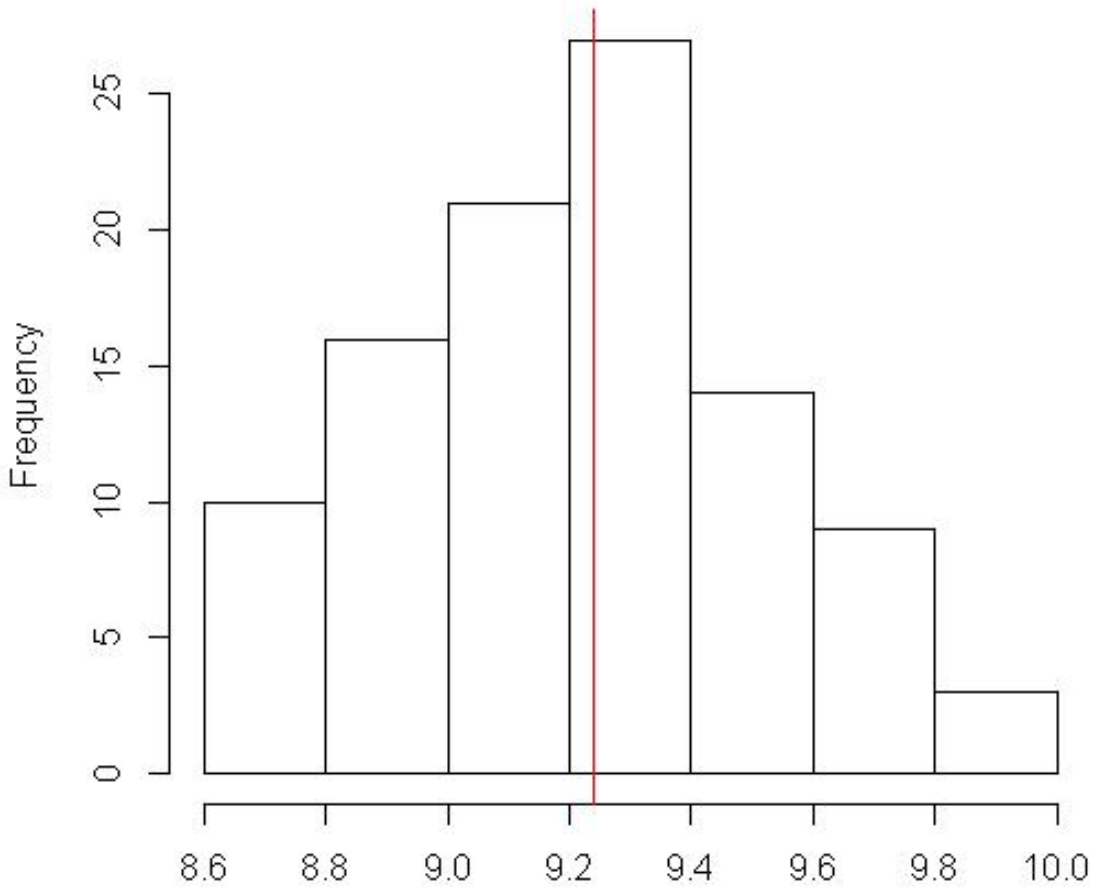

Figure: Distribution of optimal p's chosen by Pricing B: Red vertical line indicates the true optimal price 9.24

		Pricing A	Pricing B
$\beta = 0.92$	199 periods	1619.09 (26.47)	1612 (19.14)
	1000 periods	1619.09 (26.47)	1612 (19.14)
$\beta = 0.95$	199 periods	2690.35 (41.88)	2678.39 (23.49)
	1000 periods	2690.45 (41.88)	2678.49 (23.49)
$\beta = 0.98$	199 periods	6858.56 (101.33)	6834.01 (47.57)
	1000 periods	6984.78 (102.74)	6960.53 (47.66)

Table: Discounted profits of pricing scheme A and B: average (s.e.) of 100 replications

	Pricing A	Pricing B
$\beta = 0.92$		0.53 (0.39)
$\beta = 0.95$	0.72 (0.61)	0.44 (0.32)
$\beta = 0.98$		0.47 (0.38)

Table: Distance from the true coefficients of pricing A and B: average (s.e.) of 100 replications

True coefficients: (b_0, b_1, b_2)=(5.01,-0.50,-0.17)